

Supplementary Material: A Controlled Empirical Comparison of Classical and Quantum Kernel SVMs for Breast Cancer Classification

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Permanent Repository Tag: v1.0.0 (<https://github.com/SinaQP/SVM-Vs-QSVM>)

Primary Endpoint: Malignant Class F1 Score (pos_label=0)

Supplementary Section S1: Extended Sample-Size Scaling Evaluation

To evaluate model resilience across diverse training sample sizes, models were trained on nested subsets $N_{\text{train}} \in \{50, 100, 200, 300, 455\}$ with preprocessing pipelines (`StandardScaler`, `PCA`, `MinMaxScaler`) independently refit on each subset and evaluated on fixed held-out outer test partitions ($N_{\text{test}} = 114$).

Table 1: Complete Sample-Size Scaling Results across 5 Outer Splits (Mean $\pm$ SD).

N_{train}	Model Architecture	Family	Repr.	Qubits	Test Accuracy	Precision	Recall	Malignant F1	ROC-AUC
50	Linear SVM	Classical	PCA 2	0	0.9333 $\pm$ 0.0354	0.9126 $\pm$ 0.0485	0.9238 $\pm$ 0.0435	0.9157 $\pm$ 0.0402	0.9874 $\pm$ 0.0087
50	RBF SVM	Classical	PCA 2	0	0.9263 $\pm$ 0.0282	0.9201 $\pm$ 0.0321	0.8857 $\pm$ 0.0487	0.9008 $\pm$ 0.0353	0.9839 $\pm$ 0.0072
50	QSVC (reps=1, full)	Quantum	PCA 2	2	0.8246 $\pm$ 0.0387	0.8174 $\pm$ 0.0531	0.6857 $\pm$ 0.0911	0.7398 $\pm$ 0.0531	0.8850 $\pm$ 0.0484
50	Linear SVM	Classical	PCA 4	0	0.9316 $\pm$ 0.0353	0.9272 $\pm$ 0.0418	0.9048 $\pm$ 0.0538	0.9128 $\pm$ 0.0395	0.9894 $\pm$ 0.0090
50	RBF SVM	Classical	PCA 4	0	0.9421 $\pm$ 0.0295	0.9431 $\pm$ 0.0288	0.9095 $\pm$ 0.0631	0.9222 $\pm$ 0.0385	0.9864 $\pm$ 0.0060
50	QSVC (reps=1, full)	Quantum	PCA 4	4	0.7263 $\pm$ 0.0716	0.7412 $\pm$ 0.1691	0.4095 $\pm$ 0.2312	0.4994 $\pm$ 0.2110	0.7636 $\pm$ 0.1082
100	Linear SVM	Classical	PCA 2	0	0.9474 $\pm$ 0.0175	0.9341 $\pm$ 0.0255	0.9286 $\pm$ 0.0352	0.9287 $\pm$ 0.0227	0.9884 $\pm$ 0.0067
100	RBF SVM	Classical	PCA 2	0	0.9246 $\pm$ 0.0237	0.9098 $\pm$ 0.0388	0.8905 $\pm$ 0.0457	0.8966 $\pm$ 0.0317	0.9849 $\pm$ 0.0092
100	QSVC (reps=1, full)	Quantum	PCA 2	2	0.8614 $\pm$ 0.0190	0.8465 $\pm$ 0.0361	0.7619 $\pm$ 0.0435	0.7983 $\pm$ 0.0207	0.9214 $\pm$ 0.0101
100	Linear SVM	Classical	PCA 4	0	0.9421 $\pm$ 0.0288	0.9298 $\pm$ 0.0392	0.9190 $\pm$ 0.0487	0.9222 $\pm$ 0.0355	0.9890 $\pm$ 0.0074
100	RBF SVM	Classical	PCA 4	0	0.9386 $\pm$ 0.0139	0.9466 $\pm$ 0.0244	0.8905 $\pm$ 0.0388	0.9155 $\pm$ 0.0217	0.9884 $\pm$ 0.0044
100	QSVC (reps=1, full)	Quantum	PCA 4	4	0.7947 $\pm$ 0.0332	0.7958 $\pm$ 0.0631	0.6095 $\pm$ 0.1472	0.6733 $\pm$ 0.0978	0.8628 $\pm$ 0.0326
200	Linear SVM	Classical	PCA 2	0	0.9421 $\pm$ 0.0159	0.9254 $\pm$ 0.0267	0.9238 $\pm$ 0.0318	0.9234 $\pm$ 0.0213	0.9867 $\pm$ 0.0079
200	RBF SVM	Classical	PCA 2	0	0.9316 $\pm$ 0.0157	0.9132 $\pm$ 0.0288	0.9048 $\pm$ 0.0352	0.9066 $\pm$ 0.0202	0.9864 $\pm$ 0.0045
200	QSVC (reps=1, full)	Quantum	PCA 2	2	0.8877 $\pm$ 0.0144	0.8711 $\pm$ 0.0318	0.8190 $\pm$ 0.0487	0.8406 $\pm$ 0.0259	0.9453 $\pm$ 0.0115
200	Linear SVM	Classical	PCA 4	0	0.9579 $\pm$ 0.0157	0.9534 $\pm$ 0.0235	0.9333 $\pm$ 0.0318	0.9429 $\pm$ 0.0210	0.9910 $\pm$ 0.0029
200	RBF SVM	Classical	PCA 4	0	0.9509 $\pm$ 0.0159	0.9438 $\pm$ 0.0267	0.9286 $\pm$ 0.0318	0.9337 $\pm$ 0.0216	0.9907 $\pm$ 0.0047
200	QSVC (reps=1, full)	Quantum	PCA 4	4	0.8579 $\pm$ 0.0287	0.8465 $\pm$ 0.0538	0.7619 $\pm$ 0.0768	0.7943 $\pm$ 0.0440	0.9298 $\pm$ 0.0148
300	Linear SVM	Classical	PCA 2	0	0.9439 $\pm$ 0.0268	0.9278 $\pm$ 0.0388	0.9286 $\pm$ 0.0352	0.9258 $\pm$ 0.0328	0.9892 $\pm$ 0.0038
300	RBF SVM	Classical	PCA 2	0	0.9386 $\pm$ 0.0184	0.9245 $\pm$ 0.0288	0.9143 $\pm$ 0.0352	0.9168 $\pm$ 0.0238	0.9868 $\pm$ 0.0042
300	QSVC (reps=1, full)	Quantum	PCA 2	2	0.9000 $\pm$ 0.0190	0.8934 $\pm$ 0.0392	0.8333 $\pm$ 0.0435	0.8587 $\pm$ 0.0244	0.9554 $\pm$ 0.0135
300	Linear SVM	Classical	PCA 4	0	0.9649 $\pm$ 0.0115	0.9632 $\pm$ 0.0188	0.9429 $\pm$ 0.0213	0.9525 $\pm$ 0.0141	0.9934 $\pm$ 0.0031
300	RBF SVM	Classical	PCA 4	0	0.9561 $\pm$ 0.0159	0.9487 $\pm$ 0.0235	0.9333 $\pm$ 0.0318	0.9398 $\pm$ 0.0205	0.9918 $\pm$ 0.0039
300	QSVC (reps=1, full)	Quantum	PCA 4	4	0.8877 $\pm$ 0.0237	0.8654 $\pm$ 0.0487	0.8286 $\pm$ 0.0487	0.8436 $\pm$ 0.0335	0.9458 $\pm$ 0.0188
455	Linear SVM	Classical	PCA 2	0	0.9509 $\pm$ 0.0078	0.9266 $\pm$ 0.0318	0.9429 $\pm$ 0.0213	0.9341 $\pm$ 0.0093	0.9894 $\pm$ 0.0030
455	RBF SVM	Classical	PCA 2	0	0.9456 $\pm$ 0.0096	0.9245 $\pm$ 0.0182	0.9286 $\pm$ 0.0238	0.9263 $\pm$ 0.0133	0.9810 $\pm$ 0.0119
455	QSVC (reps=1, full)	Quantum	PCA 2	2	0.9070 $\pm$ 0.0237	0.9064 $\pm$ 0.0457	0.8381 $\pm$ 0.0832	0.8678 $\pm$ 0.0388	0.9644 $\pm$ 0.0244
455	Linear SVM	Classical	PCA 4	0	0.9684 $\pm$ 0.0100	0.9671 $\pm$ 0.0255	0.9476 $\pm$ 0.0391	0.9565 $\pm$ 0.0143	0.9952 $\pm$ 0.0029
455	RBF SVM	Classical	PCA 4	0	0.9561 $\pm$ 0.0139	0.9434 $\pm$ 0.0244	0.9381 $\pm$ 0.0398	0.9401 $\pm$ 0.0198	0.9935 $\pm$ 0.0045
455	QSVC (reps=1, full)	Quantum	PCA 4	4	0.9053 $\pm$ 0.0423	0.8748 $\pm$ 0.0767	0.8762 $\pm$ 0.0832	0.8721 $\pm$ 0.0556	0.9581 $\pm$ 0.0227

Supplementary Section S2: Computational Complexity and Hardware Profiling

Table 2: Execution Time, Computational Complexity, and Storage Footprint.

Pipeline Stage / Architecture	Platform	Qubits	Mean Runtime (s)	Rel. Cost	Time Complexity	Memory Complexity
Linear SVM (PCA 2)	CPU LibSVM	0	0.0048 $\pm$ 0.0007	1.0 $\times$ (Ref)	$\mathcal{O}(N_{\text{train}} \cdot d)$	$\mathcal{O}(N_{\text{train}} \cdot d)$
RBF SVM (PCA 2)	CPU LibSVM	0	0.0072 $\pm$ 0.0024	1.5 $\times$	$\mathcal{O}(N_{\text{train}}^2 \cdot d)$	$\mathcal{O}(N_{\text{train}} \cdot d)$
Linear SVM (PCA 4)	CPU LibSVM	0	0.0083 $\pm$ 0.0061	1.7 $\times$	$\mathcal{O}(N_{\text{train}} \cdot d)$	$\mathcal{O}(N_{\text{train}} \cdot d)$
RBF SVM (PCA 4)	CPU LibSVM	0	0.0071 $\pm$ 0.0019	1.5 $\times$	$\mathcal{O}(N_{\text{train}}^2 \cdot d)$	$\mathcal{O}(N_{\text{train}} \cdot d)$
QSVC (2Q Statevector)	Vectorized CPU	2	0.2257 $\pm$ 0.0232	$\sim 47\times$	$\mathcal{O}(N \cdot 2^q + N^2 \cdot 2^q)$	$\mathcal{O}(N^2)$ (2.07 MB)
QSVC (4Q Statevector)	Vectorized CPU	4	0.6599 $\pm$ 0.0863	$\sim 80\times$	$\mathcal{O}(N \cdot 2^q + N^2 \cdot 2^q)$	$\mathcal{O}(N^2)$ (2.07 MB)
<i>Historical 2Q ComputeUncompute</i>	Circuit Pair Sampler	2	120.5 $\pm$ 15.2	$\sim 25,000\times$	$\mathcal{O}(N^2)$ circuit pairs	$\mathcal{O}(N^2)$ Gram
<i>Historical 4Q ComputeUncompute</i>	Circuit Pair Sampler	4	455.0 $\pm$ 40.0	$\sim 55,000\times$	$\mathcal{O}(N^2)$ circuit pairs	$\mathcal{O}(N^2)$ Gram

Supplementary Section S3: Supplementary Diagnostic Figures

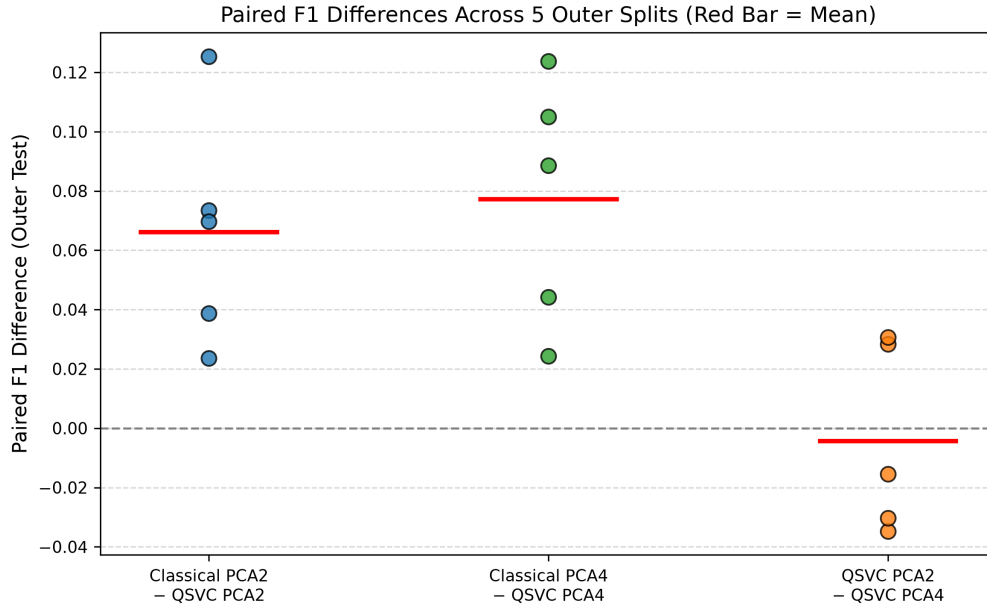


Figure 1: **Figure S1: Paired Split-Level F1 Differences.** Distribution of paired differences $\Delta = F1_{\text{Classical}} - F1_{\text{QSVC}}$ across five outer test splits. Classical models achieve higher malignant F1 scores on 5/5 splits in both PCA 2 and PCA 4 representations.

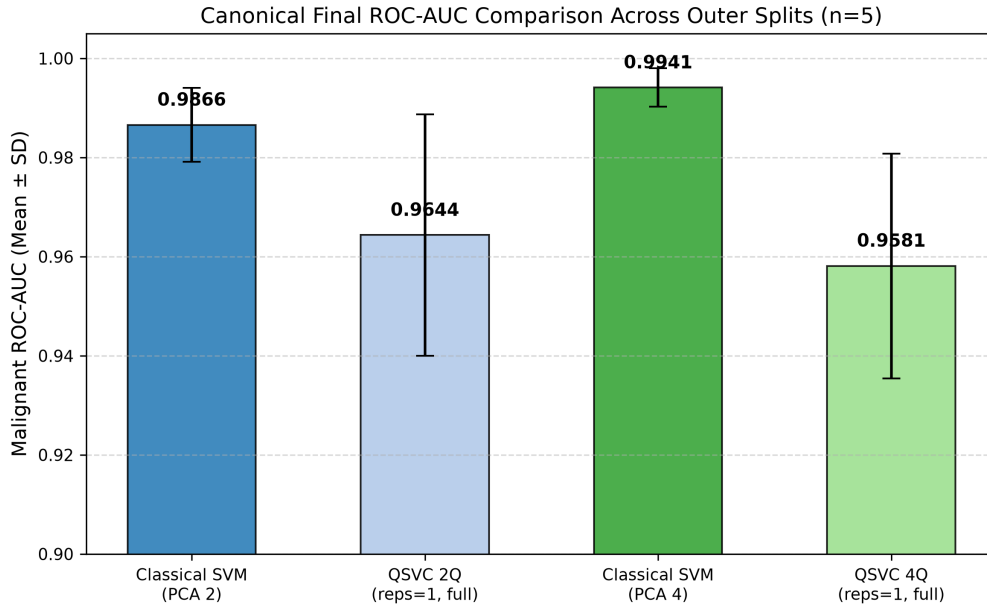


Figure 2: **Figure S2: Outer-Test ROC-AUC Comparison.** Mean receiver operating characteristic area under curve across five outer splits comparing classical and quantum models.